

Image Pyramids - Detailed Explanation

1) What is an Image Pyramid?

An Image Pyramid is a collection of images at different resolutions (sizes).

Example:

Level 0: 512x512 (Original Image)

Level 1: 256x256

Level 2: 128x128

Level 3: 64x64

2) Why Do We Use Image Pyramids?

Objects appear at different scales in images.

Example: A car near the camera looks big; far from camera looks small.

Image pyramids help detect objects at multiple scales.

3) Gaussian Pyramid

Steps:

Step 1: Apply Gaussian smoothing.

Step 2: Downsample (reduce size by half).

Repeat these steps for multiple levels.

Each level becomes smoother and smaller.

Example (Conceptual):

Original Pixel = 150

After smoothing = 140

Difference represents detail loss.

4) Laplacian Pyramid

Laplacian Level = Gaussian Level - Expanded Next Gaussian Level

It captures detail (edges and fine textures).

Gaussian pyramid stores smooth images.

Laplacian pyramid stores detailed information.

5) Why Apply Gaussian Before Downsampling?

To avoid aliasing (distortion).

Gaussian smoothing removes high-frequency noise before resizing.

6) Applications of Image Pyramids

- Object Detection
- Face Detection
- Image Blending
- Image Compression
- Feature Pyramid Networks (Deep Learning)

Simple Explanation for Students:

Image pyramid is like zooming out step by step.

Each level makes the image smaller and smoother.